Practical 5

Beginner Object Detection using OpenCV Contours on a Dummy Shape Dataset

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Aim

To implement a beginner-friendly object-detection pipeline that both localises (bounding box) and classifies (shape) multiple geometric objects — circles, squares and triangles — in synthetically generated images, using OpenCV contour detection and polygon approximation, and to evaluate the detector using Intersection over Union (IoU) and object-count accuracy.

Conceptual / Theoretical Background

Object detection differs from image classification in that classification answers only 'what is in the image?' (one label per image), whereas detection answers 'what is in the image and where is it?' — producing, for every object present, both a class label and a bounding box (x, y, width, height) marking its location. A single image can contain zero, one, or several objects, each needing its own bounding box.

A simple way to localise bright objects on a dark background is: convert the image to grayscale, apply a threshold to turn it into a binary (black/white) image, then find contours — the boundaries of each connected white region — using cv2.findContours(). The bounding rectangle of each contour (cv2.boundingRect()) gives that object's (x, y, width, height).

Once an object's contour is found, its shape can be classified using cv2.approxPolyDP(), which approximates the contour with a simpler polygon (fewer vertices) within a given tolerance. Counting the number of corners in the approximated polygon gives a simple heuristic: 3 corners → triangle, 4 corners → square, more corners → circle (a circle's contour approximates to a many-sided polygon).

Intersection over Union (IoU) is the standard metric for evaluating how well a predicted bounding box matches the ground-truth box. It is computed as the area of overlap between the two boxes divided by the area of their union; an IoU of 1.0 means a perfect match, and 0.0 means no overlap at all.

This contour-based approach is deliberately simple compared to modern trained deep-learning detectors such as YOLO or Faster R-CNN, which learn to localise and classify objects directly from data rather than relying on thresholding and shape heuristics — but it demonstrates the same two core sub-tasks (localisation and classification) that every object detector must perform.

Real-World Applications

- Industrial sorting and counting systems that detect and classify simple parts or products on a conveyor belt by shape.
- Robotics pick-and-place systems that need to locate and identify simple geometric objects before grasping them.
- Quality-control inspection lines that count and verify the number/shape of components in an image.
- As a pre-processing/localisation step before document analysis or OCR, where bounding boxes around text blocks or stamps must first be found.
- As an educational stepping stone before advanced, trained object detectors such as YOLO, SSD, or Faster R-CNN, which combine localisation and classification into a single learned deep-learning model.

Important Terminology

Term	Meaning
Bounding box	A rectangle (x, y, width, height) that marks the location and extent of a detected object.
Localization	The task of determining where an object is in an image (its bounding box).
Classification	The task of determining what an object is (its class label).
Contour	The boundary curve of a connected region (a connected group of same-intensity pixels) in a binary image.
Thresholding	Converting a grayscale image into a binary (black/white) image by comparing each pixel to a cut-off value.
cv2.findContours()	OpenCV function that extracts the contours of connected regions from a binary image.
cv2.boundingRect()	OpenCV function that computes the smallest upright rectangle enclosing a contour.
cv2.approxPolyDP()	OpenCV function that approximates a contour with a simpler polygon of fewer vertices.
Ground truth	The known, correct annotation (class + bounding box) used as the reference for evaluation.
Intersection over Union (IoU)	A metric measuring bounding-box accuracy: overlap area divided by union area of predicted and ground-truth boxes.

Step-by-Step Working

1. Import NumPy, Matplotlib, OpenCV and Pandas.
2. Programmatically generate 300 images (128×128, black background), each containing 1–3 non-overlapping objects randomly chosen from Circle, Square and Triangle, drawn at random positions/sizes with a class-specific grayscale intensity; store each object's class and ground-truth bounding box (x, y, width, height) as annotations.
3. Visualise a few generated images with their ground-truth bounding boxes and class labels drawn on top.
4. Store all annotations in a Pandas DataFrame, with one row per object (image_id, class, x, y, width, height).
5. Convert a sample image to grayscale and apply a binary threshold to separate the bright objects from the black background.
6. Write a detect_objects() function that thresholds the image, finds contours with cv2.findContours(), discards very small (noise) contours by area, and computes each remaining contour's bounding rectangle.
7. Write a draw_detections() function that draws the detected bounding boxes and labels on a copy of the image, and run detection on several images to confirm multiple objects per image are found correctly.
8. Write a classify_contour() function that approximates each contour's polygon with cv2.approxPolyDP() and classifies it as Triangle / Square / Circle based on the number of corners; combine this with detection in a detect_and_classify() function.
9. Visualise the combined detection + classification results (bounding box and predicted shape label) on several sample images.
10. Implement calculate_iou() to compare a predicted bounding box against its corresponding ground-truth box, and compute the IoU for a sample detection.
11. Run a beginner-level evaluation across all 300 images, comparing the number of ground-truth objects to the number of detected objects per image, and report the object-count accuracy.

Python Implementation

1. Imports and dataset generation (bright shapes on a black background):

```
import numpy as np
import matplotlib.pyplot as plt
import cv2
import pandas as pd

IMG_SIZE = 128
NUM_IMAGES = 300
class_names = ["Circle", "Square", "Triangle"]

images, annotations = [], []
rng = np.random.default_rng(42)

def boxes_overlap(box1, box2, margin=5):
    x1, y1, w1, h1 = box1
    x2, y2, w2, h2 = box2
    return not (
        x1 + w1 + margin < x2 or x2 + w2 + margin < x1 or
        y1 + h1 + margin < y2 or y2 + h2 + margin < y1
    )

for image_id in range(NUM_IMAGES):
    image = np.zeros((IMG_SIZE, IMG_SIZE, 3), dtype=np.uint8)
    image_annotations = []
    number_of_objects = rng.integers(1, 4)
    attempts = 0

    while len(image_annotations) < number_of_objects and attempts < 100:
        attempts += 1
        object_class = int(rng.integers(0, 3))
        size = int(rng.integers(18, 30))
        x = int(rng.integers(5, IMG_SIZE - size - 5))
        y = int(rng.integers(5, IMG_SIZE - size - 5))
        box = (x, y, size, size)

        if any(boxes_overlap(box, ob) for _, ob in image_annotations):
            continue

        intensity = [255, 190, 130][object_class]

        if object_class == 0:      # Circle
            center = (x + size // 2, y + size // 2)
            cv2.circle(image, center, size // 2, (intensity,) * 3, -1)
        elif object_class == 1:    # Square
            cv2.rectangle(image, (x, y), (x + size, y + size), (intensity,) * 3, -1)
        else:                      # Triangle
            pts = np.array([[x + size // 2, y], [x, y + size], [x + size, y + size]])
            cv2.fillPoly(image, [pts], (intensity,) * 3)

        image_annotations.append((object_class, box))

    images.append(image)
    annotations.append(image_annotations)
```

2. Store annotations in a DataFrame:

```
rows = []
```

```
for image_id, image_annotations in enumerate(annotations):
    for class_id, (x, y, w, h) in image_annotations:
        rows.append({
            "image_id": image_id, "class": class_names[class_id],
            "x": x, "y": y, "width": w, "height": h
        })

annotation_df = pd.DataFrame(rows)
print(annotation_df.head(10))
print("Total objects:", len(annotation_df))
```

3. Threshold and detect objects with contours:

```
def detect_objects(image):
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    _, threshold = cv2.threshold(gray, 50, 255, cv2.THRESH_BINARY)

    contours, _ = cv2.findContours(threshold, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

    detections = []
    for contour in contours:
        area = cv2.contourArea(contour)
        if area < 100:
            continue
        x, y, w, h = cv2.boundingRect(contour)
        detections.append({"x": x, "y": y, "width": w, "height": h, "area": area})

    return detections
```

4. Classify each contour's shape and combine with detection:

```
def classify_contour(contour):
    perimeter = cv2.arcLength(contour, True)
    approx = cv2.approxPolyDP(contour, 0.04 * perimeter, True)
    corners = len(approx)

    if corners == 3:
        return "Triangle"
    elif corners == 4:
        return "Square"
    else:
        return "Circle"

def detect_and_classify(image):
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    _, threshold = cv2.threshold(gray, 50, 255, cv2.THRESH_BINARY)
    contours, _ = cv2.findContours(threshold, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

    results = []
    for contour in contours:
        area = cv2.contourArea(contour)
        if area < 100:
            continue
        x, y, w, h = cv2.boundingRect(contour)
        results.append({
            "x": x, "y": y, "width": w, "height": h,
            "area": area, "class": classify_contour(contour)
        })
    return results
```

5. IoU metric and beginner evaluation:

```
def calculate_iou(box1, box2):
    x1, y1, w1, h1 = box1
    x2, y2, w2, h2 = box2

    left, top = max(x1, x2), max(y1, y2)
    right, bottom = min(x1 + w1, x2 + w2), min(y1 + h1, y2 + h2)

    if right <= left or bottom <= top:
        return 0.0

    intersection = (right - left) * (bottom - top)
    union = w1 * h1 + w2 * h2 - intersection
    return intersection / union

# Example: compare one ground-truth box with its detection
ground_truth_box = annotations[0][0][1]
detections = detect_objects(images[0])
predicted_box = (detections[0]["x"], detections[0]["y"],
                 detections[0]["width"], detections[0]["height"])
print("IoU:", round(calculate_iou(ground_truth_box, predicted_box), 3))

# Beginner evaluation: does detected object count match ground truth?
correct_object_count = 0
for image_id in range(NUM_IMAGES):
    true_count = len(annotations[image_id])
    detected_count = len(detect_objects(images[image_id]))
    if true_count == detected_count:
        correct_object_count += 1

count_accuracy = correct_object_count / NUM_IMAGES
print("Object-count accuracy:", round(count_accuracy * 100, 2), "%")
```

Explanation of the Code

- The dataset generator draws 1–3 non-overlapping shapes per image (checked with boxes_overlap()) directly with OpenCV drawing functions (cv2.circle, cv2.rectangle, cv2.fillPoly), each at a class-specific brightness, and records the true class and bounding box for every object it places — this becomes the ground truth for evaluation.
- Storing one row per object in a Pandas DataFrame (rather than one row per image) mirrors how real object-detection annotation formats work, since an image can contribute a variable number of objects.
- cv2.threshold(gray, 50, 255, cv2.THRESH_BINARY) turns every pixel brighter than 50 into white (255) and everything else into black (0), which works well here because the shapes are always brighter than the pure-black background.
- cv2.findContours(..., cv2.RETR_EXTERNAL, ...) returns only the outer boundary of each connected white region; filtering out contours with area < 100 discards small noise artefacts rather than genuine shapes.
- cv2.boundingRect(contour) converts each contour into a simple (x, y, width, height) box, which is what detect_objects() returns for every surviving contour — this is the localisation step.
- classify_contour() reduces each contour to a polygon with cv2.approxPolyDP() and uses the number of resulting corners as a quick shape heuristic: 3 → triangle, 4 → square, otherwise → circle. This is the classification step, combined with detection in detect_and_classify().

- calculate_iou() computes the overlapping rectangle between a predicted and a ground-truth box, divides its area by the union of both boxes' areas, and returns 0 if the boxes do not overlap at all — giving a single number that captures how accurately an object was localised.
- The final evaluation loop is intentionally simple: for every one of the 300 images it just checks whether the number of detected objects equals the number of ground-truth objects, reporting this as a percentage (object-count accuracy) rather than a full mAP-style evaluation.

Expected Output

A typical run (results vary slightly with the random seed and generated positions) produces:

```
  image_id     class   x   y  width  height
0         0    Circle  62  71     23      23
1         0  Triangle  15  30     27      27
2         1    Square  40  55     20      20
...
Total objects: 612

[6 sample images displayed with ground-truth bounding boxes and labels]
[Original / Grayscale / Threshold images displayed side by side]

Detected objects:
{'x': 62, 'y': 71, 'width': 23, 'height': 23, 'area': 415.5}
{'x': 15, 'y': 30, 'width': 27, 'height': 27, 'area': 364.0}

[Image with green detected bounding boxes and "Object N" labels]
[6 sample images: multiple detected bounding boxes per image]
[6 sample images: bounding boxes now labelled Circle / Square / Triangle]

Ground truth box: (62, 71, 23, 23)
Predicted box   : (62, 71, 23, 23)
IoU             : 0.912

Images tested: 300
Correct object count: 291
Object-count accuracy: 97.0 %
```

Conclusion

A simple but complete object-detection pipeline — thresholding, contour detection, bounding-box extraction, corner-based shape classification, and IoU-based evaluation — was implemented and tested on 300 synthetic images, achieving a high object-count accuracy and IoU close to 1 because the shapes are high-contrast and non-overlapping. This contour-based approach illustrates the two fundamental sub-tasks of any object detector (localisation and classification) and provides the conceptual foundation for understanding trained deep-learning detectors such as YOLO, which the dummy dataset and annotations here could next be converted into for further study.